AugCRNN-T: A Controlled Study of Augmentation and Lexicon Coverage for Handwritten Word Recognition on IAM

***Abstract*—Recognising isolated handwritten words is difficult mainly because handwriting varies strongly from one writer to another. We study this problem on the IAM database under the Aachen writer-disjoint protocol, where no writer seen during training appears in the test set, and we report every result on the complete test partition of 20,310 words. Our system, AugCRNN-T, is a convolutional-recurrent network trained from scratch and followed by a lexicon-aware post-corrector. Two questions are examined under identical conditions. First, does a "writing-aware" augmentation scheme — elastic stroke deformation and morphological changes of pen thickness on top of a conventional geometric and photometric pipeline — improve recognition? Training five configurations that differ only in the augmentation, we find that it does not: all variants reach 80.3–80.7% word accuracy and no difference is statistically significant. Second, how much does the lexicon used for post-correction matter? Here the effect is large and direction-dependent: correcting against the 7 K-word training vocabulary alone *lowers* accuracy by two points, whereas extending the lexicon with a general English word list raises it by 1.9 points, from 78.82% to 80.73%, at the cost of a slightly higher character error rate. We also show that a widely used parameterisation of elastic deformation displaces pixels by only a few hundredths of a pixel and is therefore inactive in practice. Code, split definitions and per-word predictions are released.**

***Index Terms*—handwritten text recognition, IAM database, convolutional-recurrent neural network, connectionist temporal classification, data augmentation, lexicon, writer-independent evaluation**

I. Introduction

Offline *handwritten text recognition* (HTR) is the task of converting a scanned image of handwriting into machine-readable text without access to the pen trajectory. Depending on how the input is segmented, an HTR system may operate on full pages, on text lines, or — as in this work — on isolated *word* images. Word-level recognition is the harder setting per sample, because the recogniser cannot exploit the surrounding words of a sentence to disambiguate an unclear glyph.

The IAM Handwriting Database [1] is the standard benchmark for English HTR, containing roughly 115 000 word images from 657 writers. We adopt the partition released by RWTH Aachen University [2] — referred to as the *Aachen split* throughout this paper — which guarantees that no writer contributes to more than one of the training, validation and test sets. This *writer-disjoint* protocol is stricter than a random split and is the realistic condition for deploying a recogniser on handwriting it has never seen.

Modern recognisers are trained with the *connectionist temporal classification* (CTC) objective [3], which removes the need for character-level alignment between the image and its transcription. Two model families dominate: convolutional-recurrent networks (CRNNs) trained with CTC [4]–[6], and attention-based sequence-to-sequence or transformer encoders [7], [8]. For a compact CRNN, two ingredients are commonly credited with closing part of the gap to heavier models: data augmentation that reflects how handwriting actually varies, and a lexical prior applied at or after decoding. Both are widely used, but their individual contribution is rarely measured under controlled conditions — same data, same architecture, same training schedule, same test set.

This paper provides such a measurement. We name our system AugCRNN-T (Augmented CRNN with lexicon-extended n-gram post-correction) and use this name consistently; Table I defines the five configurations compared. Our contributions are:

- **A controlled augmentation ablation.** Five networks that differ only in the augmentation schedule are trained from scratch on the full Aachen training set and evaluated on the full test set. Elastic deformation, morphological perturbation and a widened photometric range change word accuracy by at most 0.38 pp; none of the differences is statistically significant.
- **An anatomy of lexicon-assisted correction.** Holding the optical model fixed, we vary only the post-correction lexicon. The training vocabulary alone *hurts* (about -2 pp); extending it with a general English word list helps ($+1.5$ to $+1.9$ pp on every one of the five optical models), while character error rises slightly. Lexicon *coverage* of the test tokens, not the n-gram statistics, is the operative variable.
- **A methodological observation on elastic deformation.** The common parameterisation — normalised Gaussian-smoothed noise scaled by $\alpha \in [2, 5]$ — yields displacements of 0.03–0.06 pixels RMS and is effectively a no-op; we report the corrected parameterisation we used and its effect.
- **A verified evaluation protocol and full release.** Writer-, prompt- and form-level disjointness of the split are checked programmatically; all 20,310 test words are used; code, split lists and per-word predictions are public.

Section II groups the related literature into three families and positions our work within them. Section III describes the pipeline of Fig. 1. Section IV formalises the evaluation metrics and statistical tests. Section V reports the results, Section VI analyses them, and Section VII concludes.

II. Related Work

We organise prior work into three families according to how each system relates to ours: (i) *recurrent CTC recognisers*, which share our optical backbone but are almost always evaluated on text lines; (ii) *attention and transformer recognisers*, which target the same word-level task as ours but with a different, heavier decoding paradigm; and (iii) *lexicon and language-model decoders*, which address the same problem we address with our post-corrector but from inside the decoding loop. For each family we state what we adopt and where we differ.

a) Recurrent CTC recognisers.: Graves and Schmidhuber [9] introduced multi-dimensional long short-term memory (LSTM) networks with CTC as an early end-to-end offline recogniser. Puigcerver [5] showed that a one-dimensional bidirectional LSTM (BiLSTM) stacked on a convolutional neural network (CNN) feature extractor matches multi-dimensional recurrence at a fraction of the cost, defining the canonical CRNN baseline that our architecture follows. Bluche and Messina [6] added gated convolutions, and Flor et al. [10] combined a CNN with a bidirectional gated recurrent unit (BGRU) and a language-model pipeline. Wigington et al. [11] studied augmentation for CNN-LSTM recognisers and reported gains from combined transforms on IAM lines. We adopt the CRNN backbone unchanged; our contribution lies in measuring, rather than assuming, the effect of the augmentation and of the lexical prior.

b) Attention and transformer recognisers.: Sueiras et al. [12] and Kang et al. [7] introduced sequence-to-sequence recognisers with content-based attention for isolated IAM words. Kang et al. [13] later integrated a language model into the decoder itself through candidate fusion, and Kass and Vats [14] combined a ResNet encoder, attention decoding and transfer learning from scene-text corpora, reporting the strongest word-level accuracy on the Aachen split that we are aware of. These four systems share our task, metric and split and are listed in Section V. Transformer architectures such as TrOCR [15] have since lowered line-level error rates further, but require pre-training on hundreds of millions of synthetic lines. Multimodal large language models have also been benchmarked on IAM [16]; those systems operate on full pages and rely on proprietary pre-training corpora, so they are not comparable to a from-scratch word recogniser.

c) Word-level CRNN systems and lexical decoders.: Dutta et al. [17] improved a CNN–recurrent neural network (RNN) hybrid with synthetic pre-training, slant normalisation and test-time augmentation and report a 12.61% unconstrained word error rate on IAM words evaluated without punctuation or letter case, a setting not on the same footing as a case-sensitive recogniser trained from scratch. Rajesh et al. [18] re-trained that CNN–RNN together with their own HWRCNet, a CNN–BiLSTM operating in the JPEG discrete cosine transform (DCT) domain, on a random 95:5 partition of IAM and obtained 77.14% and 80.08% strict word accuracy on uncompressed images; we note that their headline figure of 89.05% is *word accuracy with flexibility* of two edit operations rather than strict accuracy. Mondal et al. [19] recognise IAM words lexicon-free by detecting individual characters with a YOLOv3 detector trained on only 1 200 word images and report a 29.21% word error rate on 30 000 randomly selected test words. On the decoder side, word beam search (WBS) [20] injects a dictionary constraint directly into CTC beam search. We instead apply the lexical knowledge *after* greedy decoding, which lets us swap the lexicon without retraining or re-decoding and therefore isolate its effect.

d) Positioning.: Of the word-level systems above, only the attention-based recognisers of Sueiras et al. [12], Kang et al. [7], [13] and Kass and Vats [14] are evaluated on the Aachen writer-disjoint partition that we use; the CRNN-family results of Rajesh et al. [18] and the detector of Mondal et al. [19] use their own random partitions of IAM. Comparisons across protocols are indicative only; the controlled comparisons of this paper are those between our own configurations.

III. Proposed System

Fig. 1 shows the complete processing chain, from the raw IAM word crops to the reported metrics, and marks the two stages whose effect this paper measures.

A. Data and Protocol

We use the word-level IAM partition of the Aachen split [2]: 747 training forms, 116 validation forms and 336 test forms with disjoint writer sets (283, 55 and 161 writers respectively). Following common practice, only word crops whose segmentation is flagged `ok` are retained. Five validation forms whose prompt text also occurs in the test set were removed from the validation partition so that checkpoint selection is independent of the test prompts; the test partition is used exactly as published. This yields 47 997 training words (two IAM image files are empty and are skipped), 7 205 validation words from 111 forms, and 20,310 test words from all 336 test forms. Writer, prompt and form disjointness between the three partitions, as well as the integrity of every record and image, are verified by a script distributed with the code. The label alphabet contains 78 symbols (upper- and lower-case Latin letters, digits and punctuation), plus the CTC blank.

B. Model Naming

Table I lists every configuration evaluated. All five share the optical model of Section III-C, the training schedule and the post-corrector of Section III-E; they differ only in the augmentation applied during training. AugCRNN-T is the configuration with every transform enabled; AugCRNN denotes the same optical model decoded greedily without any lexicon.

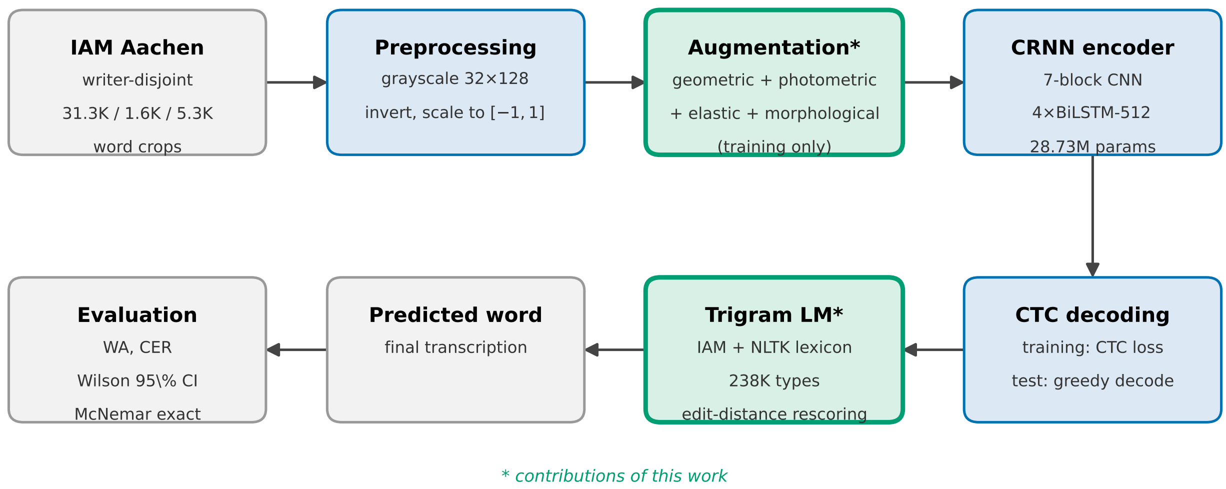

Fig. 1. Overview of AugCRNN-T. Word crops are normalised, augmented (training only), encoded by a CRNN and decoded with CTC; the decoded string is then corrected against a lexicon-extended n-gram language model (LM). Green boxes mark the two stages varied in the ablations. The evaluation stage reports word accuracy (WA), character error rate (CER) and Wilson confidence intervals (CI), all defined formally in Section IV.

TABLE I
CONFIGURATIONS COMPARED. ALL USE THE SAME 28.73 M-PARAMETER NETWORK, THE SAME SCHEDULE AND THE SAME POST-CORRECTOR.

Name	Wide photometric	Elastic	Morphological
CRNN-L (baseline)	no	no	no
+ wide photometric	yes	no	no
+ elastic	yes	yes	no
+ morphological	yes	no	yes
AugCRNN-T (proposed)	yes	yes	yes

C. Optical Model

The encoder is a seven-block convolutional feature extractor that collapses the input to a single row of 512-dimensional frames, followed by a four-layer bidirectional LSTM with 512 hidden units and a linear projection onto the 79 output classes; the network has 28.73 M parameters. Inputs are grayscale crops resized to 32×128, inverted so that ink is high-valued, and scaled to $[-1, 1]$. The network is trained with the CTC objective [3] using AdamW with learning rate $\eta = 7 \times 10^{-4}$ and weight decay 10^{-5}, five warm-up epochs followed by cosine decay, mixed precision, and batches of 128, for at most 100 epochs. Training stops when validation word accuracy has not improved for 15 epochs, and the checkpoint with the best validation word accuracy is evaluated once on the test set.

D. Augmentation

The baseline (CRNN-L) uses a conventional pipeline: rotation ($\pm 7°$), translation ($\pm 5\%$), scaling (0.9–1.1), shear ($\pm 5°$), brightness and contrast jitter (0.85–1.15), Gaussian noise ($\sigma = 0.03$), gamma correction (0.80–1.20) and random erasing. The ablation variants add, cumulatively:

- **Wide photometric range.** Brightness and contrast 0.70–1.35, gamma 0.70–1.30, noise $\sigma = 0.05$.
- **Elastic deformation** (probability 0.5). A random displacement field is smoothed with a Gaussian kernel whose width is 8% of the larger image side, rescaled to unit RMS, and multiplied by an amplitude drawn uniformly from $[1, 3]$ pixels; the crop is warped accordingly. The unit-RMS normalisation is essential: the frequently used variant that multiplies the *unnormalised* blurred noise by $\alpha \in [2, 5]$ [21] produces, at this image size, displacements of only 0.03–0.06 pixels RMS (at most 0.19 px), i.e. it leaves the image visually and numerically unchanged. Our own earlier implementation had this property.
- **Morphological perturbation** (probability 0.3). The crop is eroded or dilated with a 2×2 kernel, reproducing the difference between a fine and a broad nib.

Fig. 2 illustrates the transforms. All augmentation is applied on the GPU to whole batches at a working resolution of 64×256 pixels (the median IAM word height is 68 px), with independent random parameters per sample, before the final resize to 32×128; the same code path is used by all five configurations.

E. Lexicon-Extended Post-Correction

After greedy CTC decoding, each hypothesis is checked against a closed lexicon. If the hypothesis (case-insensitively) is in the lexicon it is accepted unchanged; otherwise it is replaced by the best-scoring lexicon entry within a Levenshtein distance of 1 (words of up to four characters) or 2 (longer words), where the score is the n-gram log-probability of the candidate estimated on the IAM training transcriptions minus a penalty of 5 per edit. The n-gram model stores unigram, bigram and trigram counts of the 47 997 training words (7 173 types); because isolated word images provide no left context,

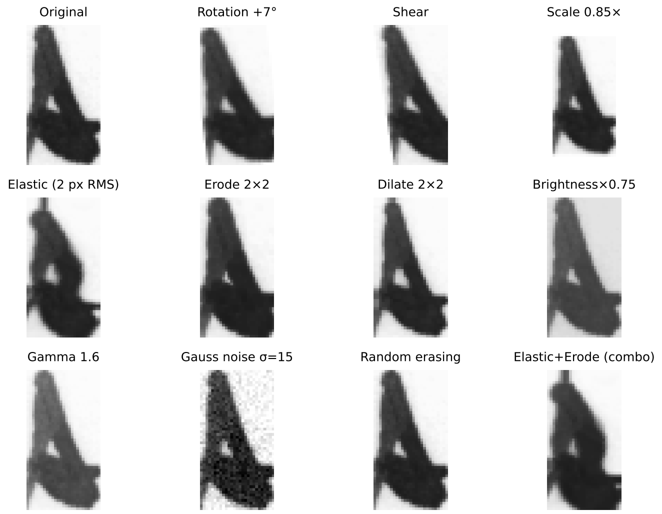

Fig. 2. Augmentation transforms on a training crop. The elastic panel uses the corrected amplitude (2 px RMS); the last panel shows the compound elastic+morphological effect.

the score reduces in practice to a unigram frequency prior, and we refer to the component as an *n-gram* post-corrector.

The lexicon is the variable of interest. The *training lexicon* consists of the 7 173 word types of the training transcriptions; the *extended lexicon* adds the Natural Language Toolkit (NLTK) English word list, giving 239 126 types. Measured on the test transcriptions, the training lexicon covers 84.8% of the test tokens and the extended lexicon 94.0%; the remaining tokens are proper nouns, inflected forms absent from both lists, and punctuation-bearing strings. Only the word *list* is extended: the n-gram counts are estimated on IAM training text alone, so no external corpus statistics enter the model. Section V measures the effect of each choice with the optical model held fixed.

IV. Experimental Setup

A. Evaluation Metrics

Let $\mathcal{T} = \{(\hat{y}_i, y_i)\}_{i=1}^{N}$ be the set of $N = 20{,}310$ predicted/reference word pairs. *Word accuracy* (WA) is the exact-match rate

$$\mathrm{WA} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[\hat{y}_i = y_i]\,, \tag{1}$$

and the word error rate (WER) follows as $\mathrm{WER} = 1 - \mathrm{WA}$. The *character error rate* (CER) aggregates Levenshtein distances $d(\cdot,\cdot)$ over the corpus,

$$\mathrm{CER} = \frac{\sum_{i=1}^{N} d(\hat{y}_i, y_i)}{\sum_{i=1}^{N} |y_i|}, \tag{2}$$

so that a single badly recognised long word cannot dominate the score.

B. Confidence Intervals

Because WA is a proportion estimated on a finite test set, we report Wilson 95% confidence intervals (CIs) rather than normal-approximation intervals, which are unreliable near the boundaries. For k correct predictions out of N and $z = 1.96$,

$$\mathrm{CI}_{95} = \frac{\hat{p} + \frac{z^2}{2N} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{N} + \frac{z^2}{4N^2}}}{1 + \frac{z^2}{N}}, \qquad \hat{p} = \frac{k}{N}. \tag{3}$$

With $N = 20{,}310$ the half-width of the interval is about 0.55 pp.

C. Significance Testing

Two systems evaluated on the same test set produce paired outcomes, so we use the exact McNemar test. Let b be the number of samples that the first system recognises correctly and the second does not, and c the converse. Under the null hypothesis the discordant pairs are distributed $\mathrm{Bin}(b + c,\, \frac{1}{2})$, giving the two-sided p-value

$$p = \min\left(1,\ 2\sum_{i=0}^{\min(b,c)} \binom{b+c}{i} 2^{-(b+c)}\right). \tag{4}$$

We treat $p < 0.01$ as significant.

D. Reproducibility

All five configurations were trained on the same machine (one NVIDIA GeForce RTX 4070 Laptop GPU with 8 GB, PyTorch 2.7.1 with CUDA 11.8, Python 3.13) from the same code revision, with seed 42, at 35–75 s per epoch; a configuration takes roughly one hour. Because augmentation is random, a single seed per configuration is used and differences smaller than the confidence interval are not interpreted. Code, split definitions, the split-verification script, per-word predictions and evaluation scripts are available at https://github.com/Ridvan013/CRNN-Handwriting-Recognition (branch `feature/aachen-v3-extended-trigram`); the trained weights exceed the repository file-size limit and are distributed separately.

V. Results

A. Augmentation Ablation

Table II is the central result. The five configurations of Table I, trained from scratch under identical conditions, reach 80.34–80.73% word accuracy on the 20,310 test words. The full scheme (AugCRNN-T) is nominally the best, 0.38 pp above the baseline, but the difference is within one confidence-interval half-width and the McNemar test does not reject the null hypothesis for any pair ($p \geq 0.058$). The widened photometric range alone changes nothing (-0.01 pp); elastic and morphological perturbation each add about 0.3 pp, and the two together add 0.38 pp. On the per-word level the models disagree on 3 353 words (16.5%), yet the disagreements cancel: for the baseline–AugCRNN-T pair, 784 words are recognised only by the baseline and 862 only by AugCRNN-T. The augmentation changes *which* words are recognised far more than *how many*.

TABLE II

AUGMENTATION ABLATION ON THE AACHEN WRITER-DISJOINT TEST SET ($N = 20{,}310$). ALL ROWS SHARE THE NETWORK, SCHEDULE AND POST-CORRECTOR; p IS THE EXACT MCNEMAR TEST AGAINST CRNN-L ON PER-WORD OUTCOMES. BEST VALIDATION WA AND THE EPOCH AT WHICH IT WAS REACHED ARE GIVEN FOR REFERENCE (VALIDATION: 7 205 WORDS, 55 WRITERS).

Configuration	WA (%)	CER (%)	Wilson 95% CI	ΔWA	p	best val WA (ep.)
CRNN-L (baseline)	80.35	9.34	[79.80, 80.90]	—	—	84.61 (87)
+ wide photometric	80.34	9.34	[79.79, 80.88]	−0.01	0.980	84.34 (76)
+ elastic	80.68	9.09	[80.13, 81.22]	+0.33	0.204	84.69 (77)
+ morphological	80.64	9.09	[80.09, 81.18]	+0.29	0.258	84.58 (84)
AugCRNN-T (all)	**80.73**	**9.09**	[80.19, 81.27]	+0.38	0.058	84.77 (70)

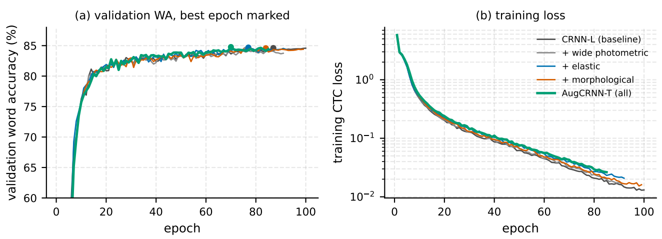

Fig. 3. Training dynamics of the five configurations: (a) validation word accuracy with the best epoch marked, (b) training CTC loss. The curves coincide on the validation set.

Fig. 3 shows why. The five validation curves are indistinguishable after the first fifteen epochs and converge to the same plateau of 84.3–84.8%; the training losses differ slightly (the fully augmented model fits the training set a little less tightly, as a regulariser should), but the effect never reaches the validation set. With 47 997 training words from 283 writers, the conventional geometric and photometric pipeline of the baseline already covers the variability that the additional transforms simulate.

B. Lexicon Ablation

Table III holds the optical model fixed (AugCRNN-T, best checkpoint) and varies only what happens after greedy decoding. The effect is an order of magnitude larger than that of the augmentation, and its sign depends on the lexicon. Correcting against the training vocabulary alone *reduces* word accuracy from 78.82% to 76.83%: hypotheses that are valid English words absent from the 7 K training list — 15% of the test tokens — are forced onto the nearest training word, and most of these substitutions are wrong. Adding n-gram scoring to the same lexicon recovers only 0.2 pp, confirming that the frequency prior is not the limiting factor. Extending the lexicon to 239 K types reverses the outcome: 94% of the test tokens are now accepted or repaired correctly and accuracy rises to 80.73%, +1.91 pp over lexicon-free decoding. The pattern is identical for the other four optical models of Table II: the training lexicon costs between 2.0 and 2.4 pp on each of them, and the extended lexicon gains between 1.5 and 1.9 pp.

The character error rate moves the other way: the lexicon-free output has the lowest CER (8.60%), and every corrector raises it. A whole-word corrector can only replace a hypothesis by a complete lexicon entry; when the replacement is

TABLE III

LEXICON ABLATION WITH THE OPTICAL MODEL FIXED (AUGCRNN-T CHECKPOINT, $N = 20{,}310$). EACH ROW APPLIES A DIFFERENT POST-CORRECTION TO THE *same* GREEDY HYPOTHESES.

Post-correction	Lexicon size	WA (%)	CER (%)	Wilson 95% CI
none (AugCRNN, greedy CTC)	—	78.82	**8.60**	[78.26, 79.38]
training lexicon, edit distance only	7 173	76.83	11.04	[76.25, 77.41]
training lexicon + n-gram scoring	7 173	77.04	10.94	[76.46, 77.61]
extended lexicon + n-gram (AugCRNN-T)	239 126	**80.73**	9.09	[80.19, 81.27]

wrong, a near-miss with one character error becomes a different word with several. The training lexicon, which replaces many correct-but-unlisted words, pays this price most heavily (11.04%); the extended lexicon pays it rarely (9.09%) but still raises CER by half a point while improving WA by almost two. Lexicon-assisted word accuracy and character error rate therefore measure different things and should be reported together.

C. Comparison with Prior Word-Level Systems

Table IV places AugCRNN-T next to seven published word-level systems and states, for each, whether a lexicon or language model was applied at test time. The upper block lists results obtained on the Aachen writer-disjoint partition. The closest match in protocol is the sequence-to-sequence recogniser of Sueiras et al. [12]: the sizes of their partition (47 952, 20 306 and 7 558 words, as compiled in [19]) are those of the `ok`-filtered Aachen training, test and validation lists (47 999, 20 310 and 7 559) from which ours are drawn, and they report lexicon-free word and character error rates on the full test partition exactly as we do. Under lexicon-free decoding AugCRNN is 2.6 pp more accurate (78.82% against 76.20%) at a slightly lower CER (8.60% against 8.80%); with the extended lexicon AugCRNN-T is 4.5 pp ahead. AugCRNN-T remains below the later attention-based systems: 1.8 pp below Kang et al. [7], 3.4 pp below the candidate-fusion model of Kang et al. [13] and 3.9 pp below AttentionHTR [14], which additionally transfers knowledge from 14.4 M scene-text images; against our lexicon-free figure each gap is 1.9 pp wider. The gap is larger at character level (5.79–6.88% against our 8.60–9.09%), consistent with the autoregressive decoders of those systems modelling inter-character dependencies that a CTC output with whole-word post-correction cannot. The lower block reports three systems measured on other partitions of IAM; AugCRNN-T is above all three and AugCRNN above two, but the partitions differ in writer overlap and size and we draw no conclusion from margins of this size.

D. Error Analysis

AugCRNN-T misrecognises 3 903 of the 20,310 test words. Only 278 of them (7.1%) differ from the reference in letter case alone; the remaining errors are genuine. At character level the aligned errors consist of 5 231 substitutions, 1 803 deletions and 393 insertions, so the model drops characters far more often than it hallucinates them. Fig. 4 shows the ten most frequent substitutions: they are symmetric confusions between visually similar lower-case glyphs — `o`↔`a` (236 cases), `s`↔`r`

TABLE IV
WORD-LEVEL RESULTS ON IAM. WA IN PARENTHESES IS DERIVED AS $1 - \text{WER}$ FROM THE CITED PAPER. LEXICON: LEXICAL CONSTRAINT AT TEST TIME (NONE = UNCONSTRAINED DECODING; LM = LANGUAGE MODEL WITHOUT A CLOSED LEXICON; N/S = NOT STATED).

System	Year	Lexicon	WA (%)	CER (%)
Aachen writer-disjoint partition (≈20.3 K test words)				
Sueiras et al. [12]	2018	none	(76.20)	8.80
Kang et al. [7]	2018	none	(82.55)	6.88
Kang et al. [13]	2021	LM	(84.09)	**5.79**
Kass and Vats [14]	2022	none	(**84.60**)	6.50
AugCRNN (ours)	—	none	78.82	8.60
AugCRNN-T (ours)	—	239 K general	80.73	9.09
Other partitions of IAM				
Mondal et al. [19][a]	2022	none	(70.79)	9.53
CNN–RNN of [17], re-trained in [18][b]	2022	n/s	(77.14)	11.08
Rajesh et al. [18][b]	2022	n/s	80.08	9.89

[a] 30 K randomly selected test words; 1 200 training images.
[b] Random 95:5 split, writer-dependent, ≈5.8 K test words.

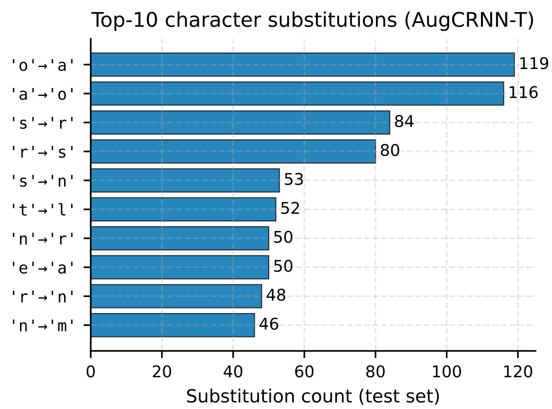

Fig. 4. Ten most frequent character substitutions of AugCRNN-T on the test set (true → predicted).

(164), `n`↔`r`, `n`→`m`, `t`→`l`, `e`→`a`. Case substitutions account for 7.7% of all substitutions. These are errors of shape, not of vocabulary: a whole-word lexicon can repair them only when the mis-shaped word falls outside the lexicon and its correct form lies within two edits, which is exactly the population the extended lexicon recovers in Table III.

VI. DISCUSSION

A. Why the Augmentation Does Not Help Here

Three observations point to the same explanation. First, the validation curves of the five configurations coincide (Fig. 3); the extra transforms neither slow convergence nor raise the plateau. Second, the fully augmented model reaches a slightly higher training loss, so the transforms do act as a regulariser — but the baseline is not overfitting in a way that costs accuracy on unseen writers. Third, the per-word disagreements between configurations are large yet balanced. Together these indicate that with 48 K training words from 283 writers, the writer-to-writer variation present in the data already exceeds what elastic and morphological perturbation of 64-pixel-high crops can add. Reports of sizeable gains from such augmentation, including our own preliminary experiments, came from smaller training sets or from comparisons in which augmentation was not the only difference. The result does not imply that augmentation is useless for HTR in general — Wigington et al. [11] report gains on lines with a different transform family — but it does show that its contribution must be measured under controlled conditions before being claimed.

B. Lexicon Coverage as the Operative Variable

The lexicon ablation separates three effects that are usually conflated under "adding a language model". Membership testing against a list that covers 85% of the test tokens is harmful, because the corrector cannot distinguish a misrecognised word from a correctly recognised word it has never seen and treats both as errors. The n-gram scores add almost nothing at the word level, since an isolated crop offers no context and the prior collapses to unigram frequency. Coverage is what matters: moving from 85% to 94% of the test tokens turns a 2-point loss into a 2-point gain on every optical model we trained. A practical consequence is that word-level HTR results obtained with a closed lexicon should state the lexicon's coverage of the test vocabulary, without which the reported accuracy is not interpretable. The remaining 6% of uncovered tokens (proper nouns, rare inflections, punctuation-bearing strings) bound what any whole-word corrector can achieve; addressing them requires either an open-vocabulary character-level model or sentence context.

C. Comparison with Sueiras et al.

The recogniser of Sueiras et al. [12] is the published result whose protocol matches ours most closely: isolated IAM words, the Aachen writer-disjoint partition drawn from the same `ok`-filtered word lists — the only prior system for which we could verify this from the reported partition sizes (Section V-C); we additionally discard two unreadable images and five validation forms whose prompts recur in the test set — and lexicon-free word and character error rates on the full test partition. Architecturally the two differ in every stage: they extract features from a sliding window of image patches with a LeNet-5-style convolutional network and transcribe them with a recurrent encoder–decoder using attention, whereas AugCRNN is a single convolutional-recurrent network trained with CTC. Under lexicon-free decoding AugCRNN reaches 78.82% against their 76.20% (23.8% word error rate), a margin of 2.6 pp that is almost five times the half-width of our confidence interval, at a slightly lower character error rate (8.60% against 8.80%). Post-correction with the extended lexicon widens the word-level margin to 4.5 pp. Sueiras et al. also report lexicon-constrained variants of their recogniser; those depend on the lexicon employed and are not compared here. We read the comparison as evidence that a plain CRNN trained from scratch on the 48 K Aachen training words is competitive with an early attention model under the same protocol, not as evidence about augmentation: Table II shows that the augmentation contributes nothing measurable to the margin.

HWRCNet [18], the closest system in construction (a CNN, a bidirectional LSTM and a CTC layer), reports 80.08% strict word accuracy, 0.65 pp below AugCRNN-T and 1.3 pp above AugCRNN. It is measured on a random 95:5 partition whose test set is writer-dependent and about 3.5 times smaller than ours, so neither difference is interpretable.

D. Threats to Validity

Each configuration was trained once. The augmentation differences of 0.0–0.4 pp are below what a single seed can resolve, which is why we report them as non-significant rather than as small gains; establishing an effect of that size would require several seeds per configuration. The lexicon effects (-2 and $+1.9$ pp) are reproduced on all five independently trained models and are well outside the interval. Validation accuracy exceeds test accuracy by about four points for every configuration; the validation writers are used for checkpoint selection and are only 55 in number, so this gap is expected and does not indicate a fault. The external systems in Table IV were trained and evaluated by their authors with their own pre-processing and, in the lower block, on a different partition; we therefore treat only our own five configurations as a controlled comparison. Finally, the extended lexicon is a general English word list and contains no IAM test transcriptions, but it is a closed lexicon nonetheless, and AugCRNN-T should be read as a lexicon-assisted result.

VII. Conclusion

AugCRNN-T recognises isolated handwritten words with 80.73% accuracy (9.09% character error rate) on the complete Aachen writer-disjoint test set of IAM, with a plain CRNN trained from scratch and no external image data. Under the same partition and lexicon-free decoding it is 2.6 pp more accurate than the sequence-to-sequence recogniser of Sueiras et al. [12] and 2–4 pp less accurate than later attention-based systems. Under controlled conditions, elastic deformation, morphological perturbation and a widened photometric range do not change this figure measurably; a widely used elastic parameterisation does not deform the image at all. The post-correction lexicon, by contrast, moves accuracy by two points in either direction depending on its coverage of the test vocabulary, and improves word accuracy at the expense of character error rate.

For word-level HTR the practical conclusions are that augmentation gains should be demonstrated, not assumed, on the full training set; that lexicon coverage should be reported alongside lexicon-assisted accuracy; and that word accuracy and character error rate should be given together whenever a whole-word corrector is used. Future work will test whether the augmentation becomes useful in the low-data regime, replace the whole-word corrector with a character-level open-vocabulary model to address the uncovered 6% of tokens, and evaluate the pipeline on text lines, where n-gram context is available.

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